

Name: _____

TOPIC TEST**Trigonometric functions (Advanced)**

- Time allowed: 45 minutes
- Part A: 10 multiple-choice questions (10 marks)
- Part B: 8 free-response questions (40 marks)
- Total: 50 marks

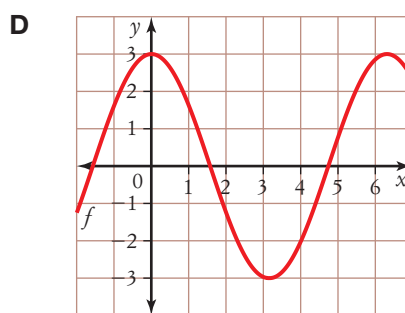
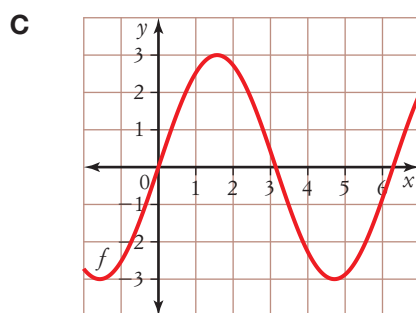
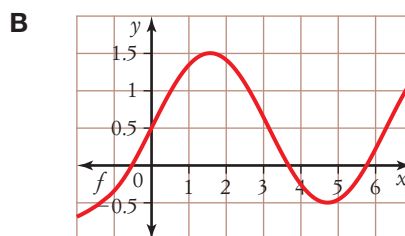
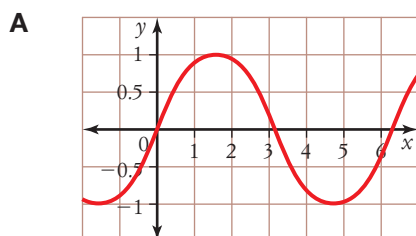
Part A

10 multiple-choice questions

1 mark each: 10 marks

Circle the correct answer.

- 1 Which graph shows a vertical dilation of $y = \sin x$ with a scale factor of 3?



- 2 Compared to $y = \sin x$, which statement is correct about $y = \sin\left(x - \frac{\pi}{2}\right) + 3$?

A The phase shift is 3 units to the right, the centre is $\frac{\pi}{2}$

B The phase shift is 3 units to the left, the centre is $\frac{\pi}{2}$

C The phase shift is $\frac{\pi}{2}$ units to the left, the centre is -3

D The phase shift is $\frac{\pi}{2}$ units to the right, the centre is 3

- 3 Which transformation will change the amplitude of a cosine function?
- A** Vertical translation **B** Vertical dilation
C Horizontal dilation **D** Horizontal translation
- 4 What is the equation of the transformed function of $y = \cos x$ with amplitude 5, period 3π , centre -1 and phase shift 3 units to the left?
- A** $-5 \cos [3(x + 3)]$ **B** $5 \cos \left[\frac{1}{3}(x + 3) \right] + 1$
C $-5 \cos [3(x + 3)]$ **D** $5 \cos \left[\frac{2}{3}(x + 3) \right] - 1$
- 5 What is the equation of the transformed function of $y = \cot x$ with period 2π and phase shift $\frac{\pi}{6}$ units to the right?
- A** $y = \cot \left(\frac{x}{2} - \frac{\pi}{6} \right)$ **B** $y = 2 \cot \left(\frac{x}{2} - \frac{\pi}{6} \right)$
C $y = \cot \left(\frac{x}{2} + \frac{\pi}{6} \right)$ **D** $y = 2 \cot \left(\frac{x}{2} + \frac{\pi}{6} \right)$
- 6 What is the equation of the transformed function of $y = \sec x$ reflected in the x -axis, with amplitude 5, period π and centre 2?
- A** $y = -5 \sec x + 2$ **B** $y = 5 \sec 2x + 2$
C $y = -5 \sec 2x + 2$ **D** $y = 5 \sec x + 2$
- 7 By considering the graph of $y = 2 \sin 3x$, what is the number of solutions for the equation $2 \sin 3x = \sqrt{2}$ in the domain $[0, 2\pi]$?
- A** 4 **B** 5
C 6 **D** 7
- 8 What is the amplitude and period of $y = -2 \sin 5x$?
- A** Amplitude 2 and period 5
B Amplitude 2 and period $\frac{2\pi}{5}$
C Amplitude -2 and period 5
D Amplitude -2 and period $\frac{2\pi}{5}$
- 9 Which function below shows a phase shift of 30° to the right and a centre of 7, compared to the parent trigonometric function?
- A** $\tan [2(x + 7)] - 30^\circ$ **B** $\cos (x + 30^\circ) + 7$
C $\sin (2x - 30^\circ) + 7$ **D** $\cos (x - 30^\circ) + 7$
- 10 Solve $\sin 3x = 1$ for the domain $[0, 2\pi]$.
- A** $x = \frac{\pi}{3}, \frac{2\pi}{3}$ **B** $x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$
C $x = \frac{\pi}{6}, \frac{11\pi}{6}$ **D** $x = \frac{\pi}{6}, \frac{5\pi}{6}$

Part B

8 free-response questions

40 marks

Show your working where appropriate.

11 Sketch the graph of each function in the domain $[0, 2\pi]$.

a $y = 3 \cos x$

b $y = \sin(-x)$

c $y = \tan(x - \pi)$

[6 marks]

12 Sketch the graph of each function in the domain $[-\pi, \pi]$.

a $y = -\sin 2x + 5$

b $y = 3 \cos[-(x + 2)]$

[4 marks]

13 Solve each trigonometric equation for the domain $[0, 2\pi]$.

a $2 \cos 2x = 1$

b $\sin\left(3x - \frac{\pi}{3}\right) = \frac{1}{\sqrt{2}}$

[6 marks]

14 Solve each trigonometric equation graphically in the domain $[0^\circ, 360^\circ]$.

a $-\sin 2x + 1 = x + 1$

b $2 \tan (x + 1) = 1$

[6 marks]

15 Solve each trigonometric equation in the domain $[0^\circ, 360^\circ]$.

a $4 \sin^2 x - 3 = 0$

b $\tan^2 2x = 1$

c $\tan^2 x - 5 \tan x + 6 = 0$

[9 marks]

16 Find the domain and range of each function.

a $y = -2 \cos x$

b $y = 5 \sin (-3x)$

[4 marks]

17 Find the amplitude and period of $y = -2 \sin (3x)$.

[2 marks]

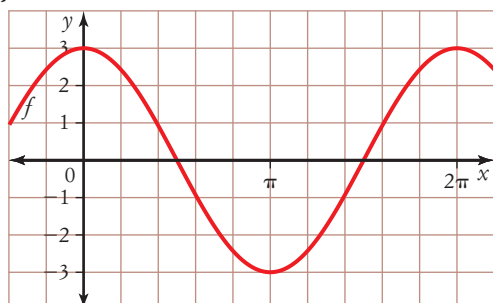
18 Show that $y = \cos x$ is the function $y = \sin x$ with a phase shift of $\frac{\pi}{2}$ to the left.

[3 marks]

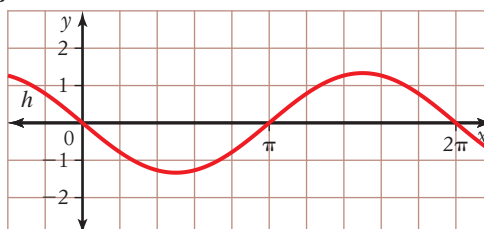
This is the end of the test.
Use this page and the back for extra working space.

Answers**1** C**2** D**3** B**4** D**5** A**6** C**7** C**8** B**9** D**10** B

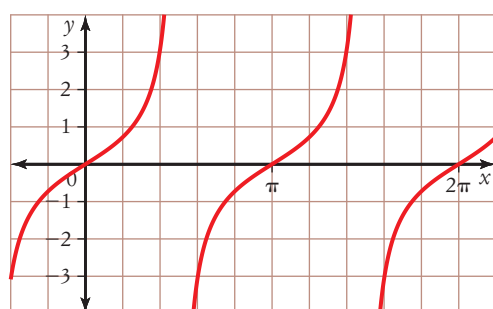
11 a $y = 3 \cos x$



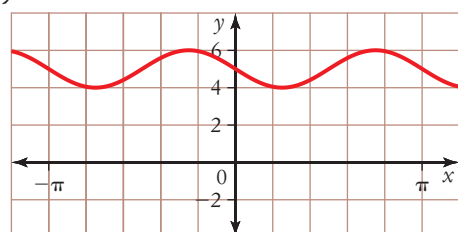
b $y = \sin(-x)$



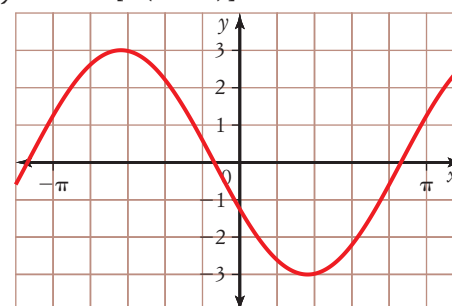
c $y = \tan(x - \pi)$



12 a $y = -\sin 2x + 5$



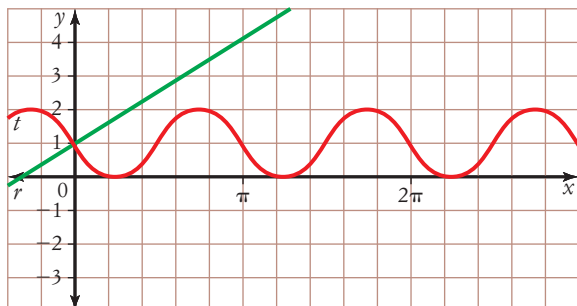
b $y = 3 \cos[-(x + 2)]$



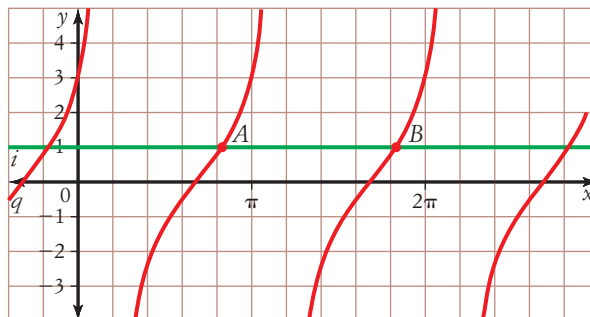
13 a $x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$

b $x = \frac{7\pi}{36}, \frac{13\pi}{36}, \frac{31\pi}{36}, \frac{37\pi}{36}, \frac{56\pi}{36}, \frac{61\pi}{36}$

14 a $x = 0, y = 1$



b $x = 149.5^\circ$ or $x = 329.5^\circ$



15 a $x = 60^\circ, 120^\circ, 240^\circ, 300^\circ$

b $x = 22.5^\circ, 67.5^\circ, 112.5^\circ, 157.5^\circ, 202.5^\circ, 247.5^\circ, 292.5^\circ, 337.5^\circ$

c $x \approx 63^\circ, 72^\circ, 243^\circ, 252^\circ$

16 a Domain: $(-\infty, \infty)$, Range: $[-1, 1]$

b Domain: $(-\infty, \infty)$, Range: $[-5, 5]$

17 a $2, \pi$

18 Teacher to check